SITHUM MANUSHA

SOFTWARE ENGINEERING INTERN

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ABOUT ME

Third-year undergraduate studying Information Technology and Management at the University of Moratuwa, concurrently reading an LL.B. at OUSL. Experienced in building full-stack web applications and backend services using React, Next.js, Java Spring Boot, and modern PHP (MVC). Focused on clean architecture, reliable database design, and practical Generative AI integrations. Seeking a Software Engineering Internship to contribute to production software systems and develop within a professional engineering team.

EDUCATION

University of Moratuwa (UOM) - Moratuwa, Sri Lanka **2024 - Present**
B.Sc. (Hons) in Information Technology & Management **(3rd Year | CGPA: 3.23 / 4.0)**

Open University of Sri Lanka (OUSL) - Nawala, Sri Lanka **2025 - Present**
Bachelor of Laws Honours Degree Programme (LL.B.)

St. Joseph's Vaz College - Wennappuwa, Sri Lanka **2009 - 2022**
G.C.E. Advanced Level (2 A's & 1 B) | G.C.E. Ordinary Level (8 A's & 1 B)

FEATURED PROJECTS

1. Super POS — Multi-Branch Enterprise Offline-First Retail System (Group Project) **2026**

- Architected a client-side offline sync engine with **IndexedDB queuing** that buffers sales and inventory changes during network outages and automatically pushes updates to **PostgreSQL** upon reconnection.
- Implemented automated **Last-Write-Wins (LWW)** conflict resolution with an interactive 3-way resolver, alongside live node health monitoring and broadcast updates using **WebSockets (Socket.io)**.
- **Technologies:** Next.js, React, TypeScript, NestJS, Node.js, PostgreSQL, Prisma ORM, Socket.io, IndexedDB, Tailwind CSS
- **Live Demo:** pos-system-pos-web-app.vercel.app | **GitHub:** github.com/Ryzera/Ryzera-POS

2. CareerFlow — Full-Stack ATS Resume Optimizer & AI Interview Platform (Individual Project) **2026**

- Developed an automated ATS resume scanner that evaluates job description alignment and identifies missing keywords using **TF-IDF vectorization and Cosine Similarity (Scikit-learn)**.
- Engineered an interactive mock interview simulator powered by a **Java 21 Spring Boot 3** REST gateway and **Python FastAPI** microservice, containerized with **Docker Compose**.
- **Technologies:** Java 21, Spring Boot 3, Python 3.12, FastAPI, Next.js 14, PostgreSQL, Docker Compose, Scikit-learn, JUnit 5
- **Live Demo:** careerflow-pink.vercel.app | **GitHub:** github.com/SithumManusha/careerflow

3. Ledgerly — Personal Finance & Shared Expense Platform (Individual Project) **2026**

- Built a personal finance tracking web application that tracks monthly expenses, computes budget category health scores, and visualizes future balance trends using **Recharts**.
- Implemented multi-member expense splitting across 4 distribution methods with automated PDF statement downloads, backed by **React, TypeScript, Express, PostgreSQL (Drizzle ORM)**, and 60+ **Vitest** unit tests.
- **Technologies:** React, TypeScript, Node.js, Express, PostgreSQL, Drizzle ORM, Recharts, Vitest, Render
- **Live Demo:** ledgerly-nine-rust.vercel.app | **GitHub:** github.com/SithumManusha/Ledgerly

4. Pulse AI Blog Engine — AI Content Platform & MVC Architecture (Individual Project) **2026**

- Architected a content publishing platform in pure **PHP 8 MVC** (Front Controller pattern) integrated with **Google Gemini 1.5 Flash API** for 1-click TL;DR summaries, auto-drafting, and tag taxonomy.
- Integrated in-browser **Web Speech API** audio playback and hardened application security using **PDO prepared statements** against SQL injection, session **CSRF tokens**, and automated **GitHub CodeQL** scans.
- **Technologies:** PHP 8, MySQL, SQLite (PDO), Google Gemini AI, Docker, GitHub Actions, CodeQL, Tailwind CSS, Web Speech API
- **Live Demo:** pulse-ai-blog-engine.onrender.com | **GitHub:** github.com/SithumManusha/pulse-ai-blog-engine

5. Automated Robotic Sorting System — Microcontroller Conveyor Platform (Group Project) **2025**

- Constructed an automated conveyor sorting platform with an Arduino-driven 4-DOF robotic arm, ultrasonic sensors, and an RFID reader to classify and distribute objects into 3 target bays.
- Calibrated stepper motor step counts and sensor timing routines to achieve 98% pick-and-place precision synchronized with continuous conveyor belt movement.
- **Technologies:** Arduino Mega/Uno (C++), Stepper Motors (NEMA 17), Servo Motors, RFID, Sensors, Integration Testing
- **GitHub:** github.com/SithumManusha/robotic-sorting-system

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, JavaScript, TypeScript, PHP, C/C++, SQL
- **Front-end Development:** React, Next.js, TypeScript, Tailwind CSS, Recharts, IndexedDB
- **Back-end Development:** Spring Boot 3, FastAPI, Node.js, Express, NestJS, PHP (MVC), WebSockets (Socket.io), RESTful APIs
- **Databases & ORM:** PostgreSQL (Supabase), MySQL, SQLite, Spring Data JPA, Hibernate, Prisma ORM, Drizzle ORM, PDO
- **DevOps, Testing & Quality:** Docker, Docker Compose, Git, GitHub Actions (CI/CD), CodeQL, JUnit 5, Pytest, Vitest, Playwright, Postman

CORE COMPETENCIES & ARCHITECTURE

MVC & Microservices Architecture, RESTful API Design, Agile / Scrum, Object-Oriented Design (SOLID), Clean Code, Automated Testing, Offline-First Sync

CERTIFICATIONS

- AWS Cloud Practitioner Essentials – Amazon Web Services (AWS)
- Software Testing with Generative AI – Simplilearn
- Introduction to Artificial Intelligence – Simplilearn | Introduction to Cybersecurity – Cisco Networking Academy

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Member of the Organizing Committee, Official Career Fair conducted by University of Moratuwa (2026)
- Member of the Organizing Committee, Project "Rest with Hope" conducted by Leo Club, UoM (2025)
- Active Member – FIT Moments (Faculty Media Body) | Leo Club of UoM | Legal Youth Community Sri Lanka

LANGUAGES

- English: Professional Working Proficiency (Oral & Written) | Sinhala: Native Proficiency

REFERENCES

Dr. A.L.A.R.R. Thanuja (B.Sc. (Sri Lanka), Ph.D. (USA)),
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